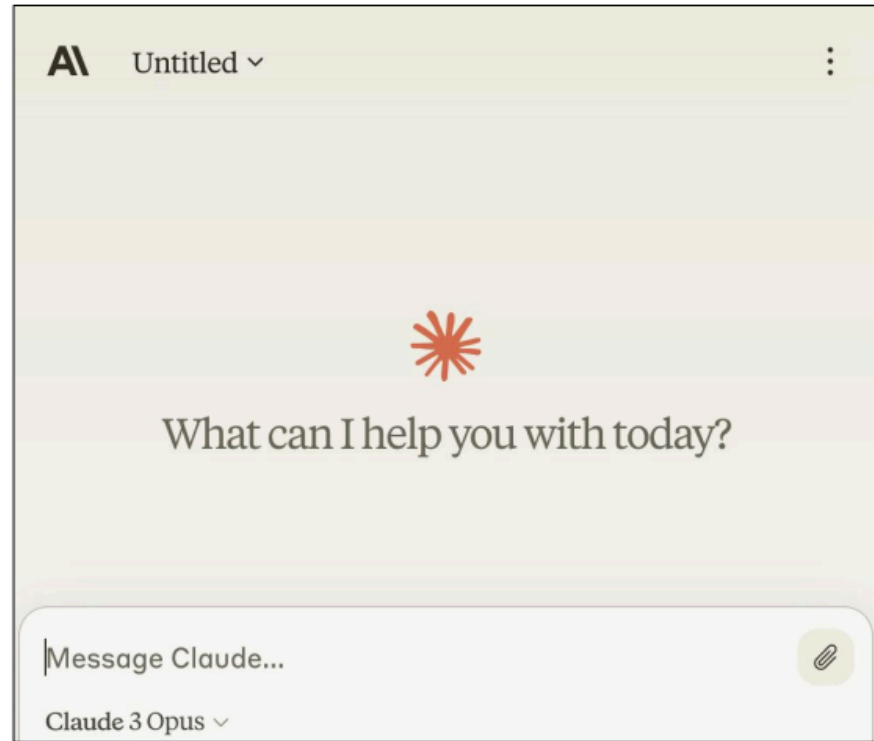
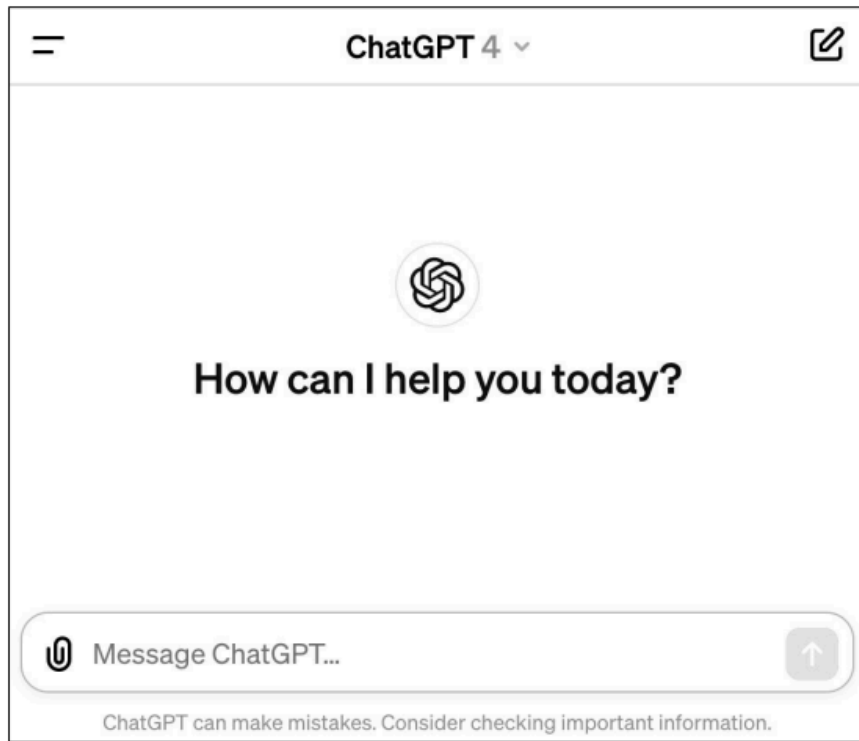


Artificial Intelligence

Lecture10 – LLMs & Transformers



- Feature engineering

- Text tokenization
- Word embeddings

- Deep neural networks

- Autoregressive models
- Self-attention mechanisms
- Transformer architecture

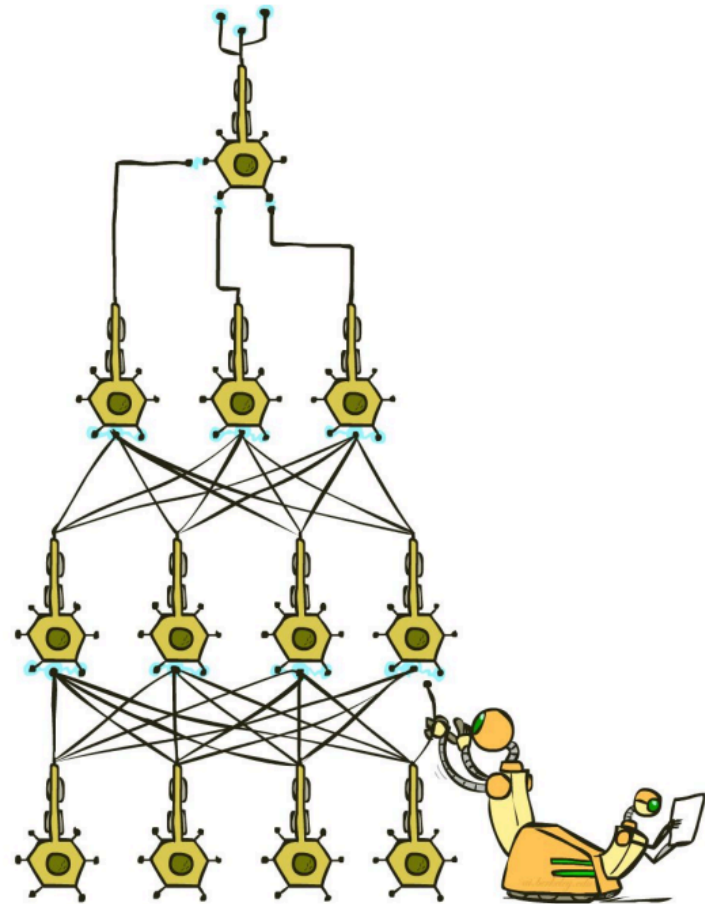
- Multi-class classification

- Supervised learning

- Self-supervised learning
- Instruction tuning

- Reinforcement learning

- ... from human feedback (RLHF)



- Input: some text
 - “The dog chased the”

- Output: more text

... “ball”

- Implementation:
 - Linear algebra
 - How??

GPT-3.5 & GPT-4

GPT-3 (Legacy)

Many words map to one token, but some don't: indivisible.

Unicode characters like emojis may be split into many tokens containing the underlying bytes: 🖐

Sequences of characters commonly found next to each other may be grouped together: 1234567890

Clear

Show example

Tokens

Characters

57

252

GPT-3.5 & GPT-4

GPT-3 (Legacy)

Many words map to one token, but some don't: indivisible.

Unicode characters like emojis may be split into many tokens containing the underlying bytes: 🍌🍌🍌🍌🍌🍌

Sequences of characters commonly found next to each other may be grouped together: 1234567890

Text

Token IDs

Tokens

Characters

57

252

GPT-3.5 & GPT-4

GPT-3 (Legacy)

```
[8607, 4339, 2472, 311, 832, 4037, 11, 719, 1063, 1541, 956, 25, 3687,
23936, 382, 35020, 5885, 1093, 100166, 1253, 387, 6859, 1139, 1690,
11460, 8649, 279, 16940, 5943, 25, 11410, 97, 248, 9468, 237, 122, 271,
1542, 45045, 315, 5885, 17037, 1766, 1828, 311, 1855, 1023, 1253, 387,
41141, 3871, 25, 220, 4513, 10961, 16474, 15]
```

Text

Token IDs

Tokens

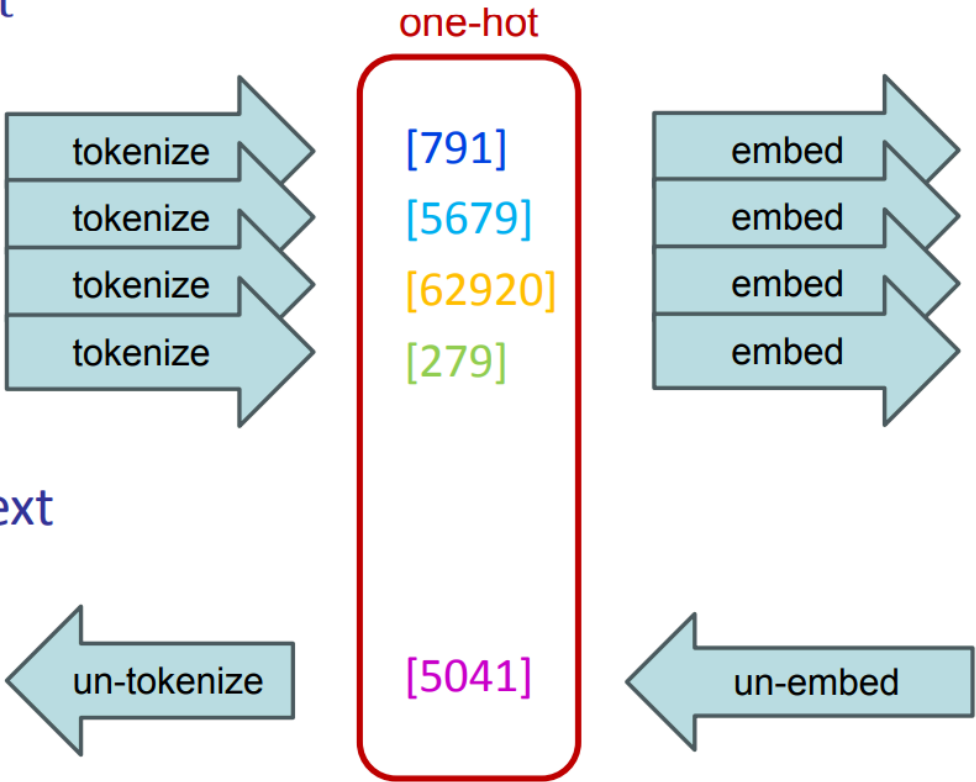
57

Characters

252

■ Input: some text

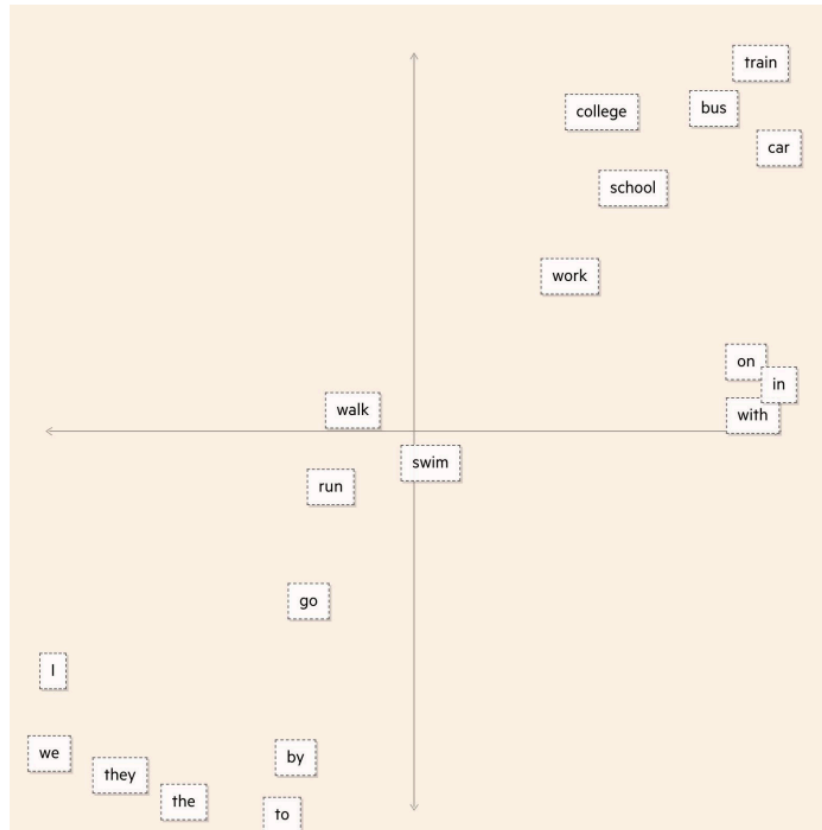
- “The”
- “ dog”
- “ chased”
- “ the”



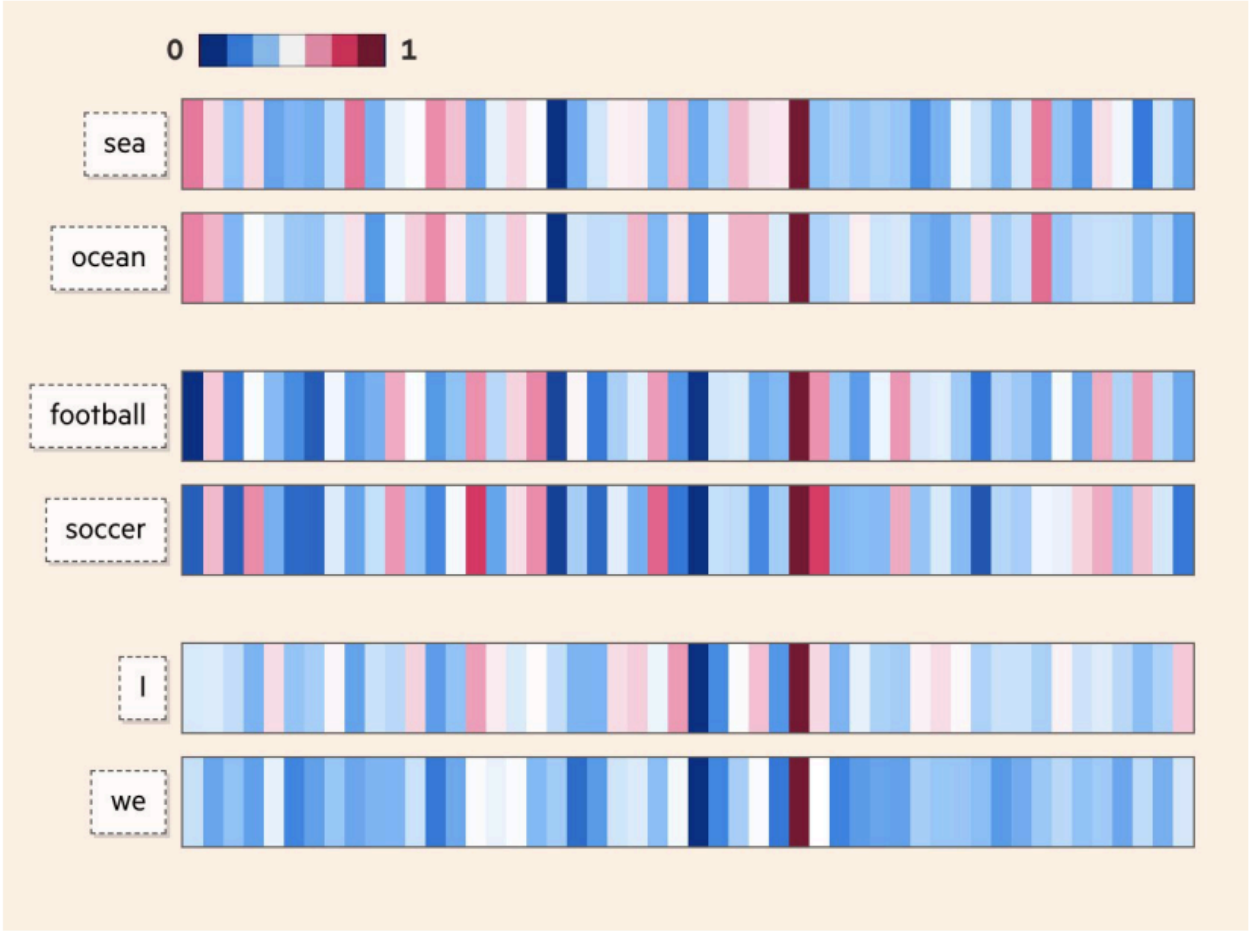
■ Output: more text

- “ ball”

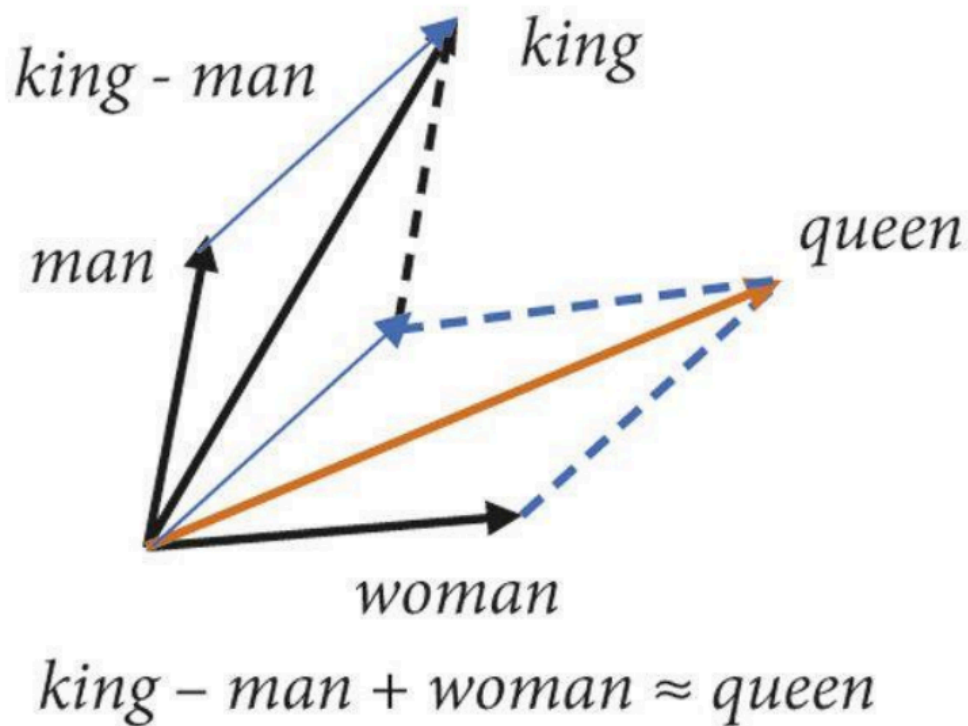
- Words cluster by similarity:



- Features learned in language models:



- Signs of sensible algebra in embedding space:



[Efficient estimation of word representations in vector space, Mikolov et al, 2013]

Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings

Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, Adam Kalai

Artificial Intelligence

Generative AI exists because of the transformer

This is how it works

By Visual Storytelling Team and Madhumita Murgia in London SEPTEMBER 11 2023

- ~~Feature engineering~~

- ~~Text tokenization~~
- ~~Word embeddings~~

- Deep neural networks

- Autoregressive models
- Self-attention mechanisms
- Transformer architectures

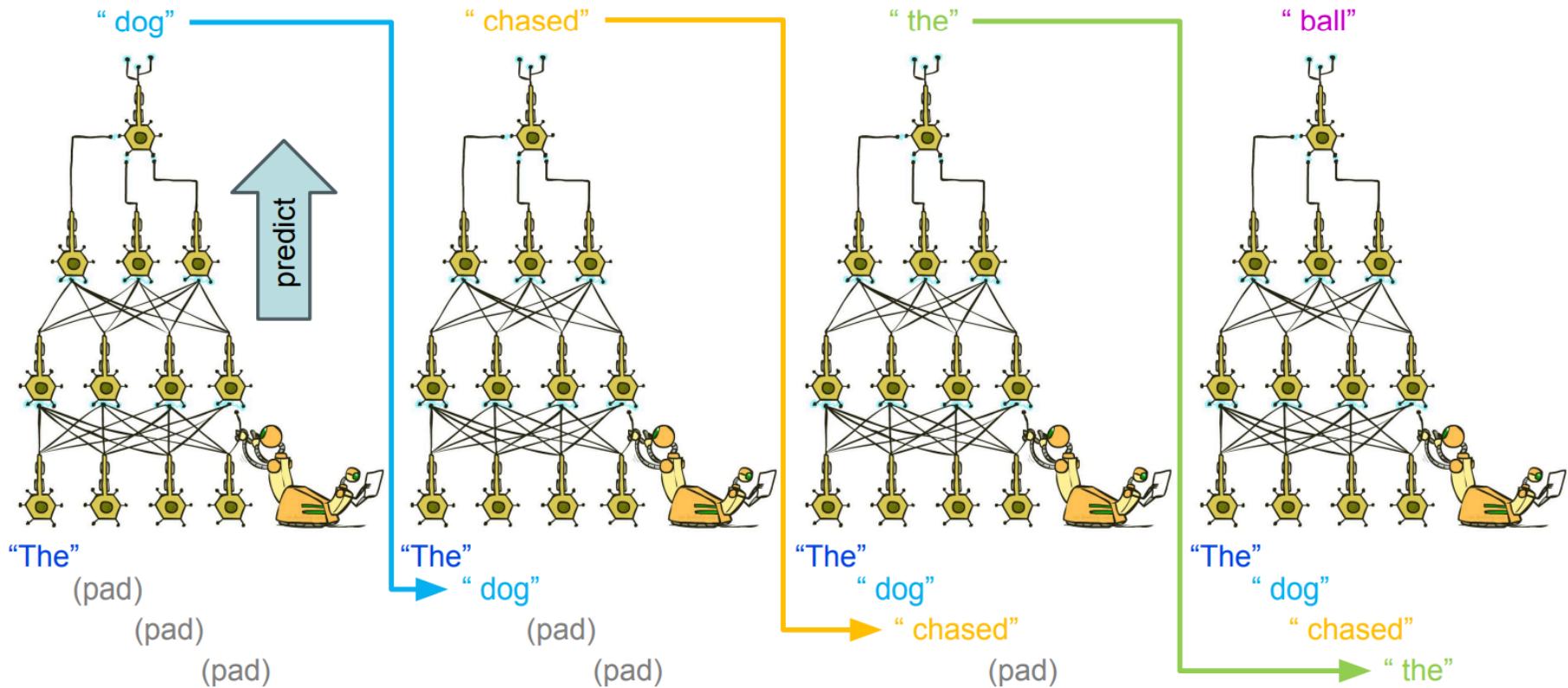
- Multi-class classification

- Supervised learning

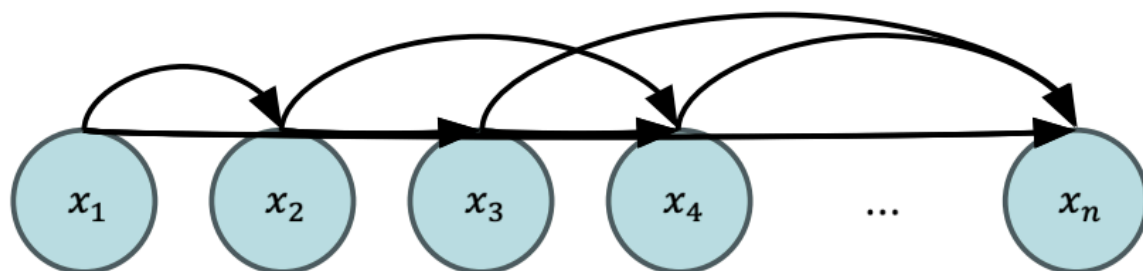
- Self-supervised learning
- Instruction tuning

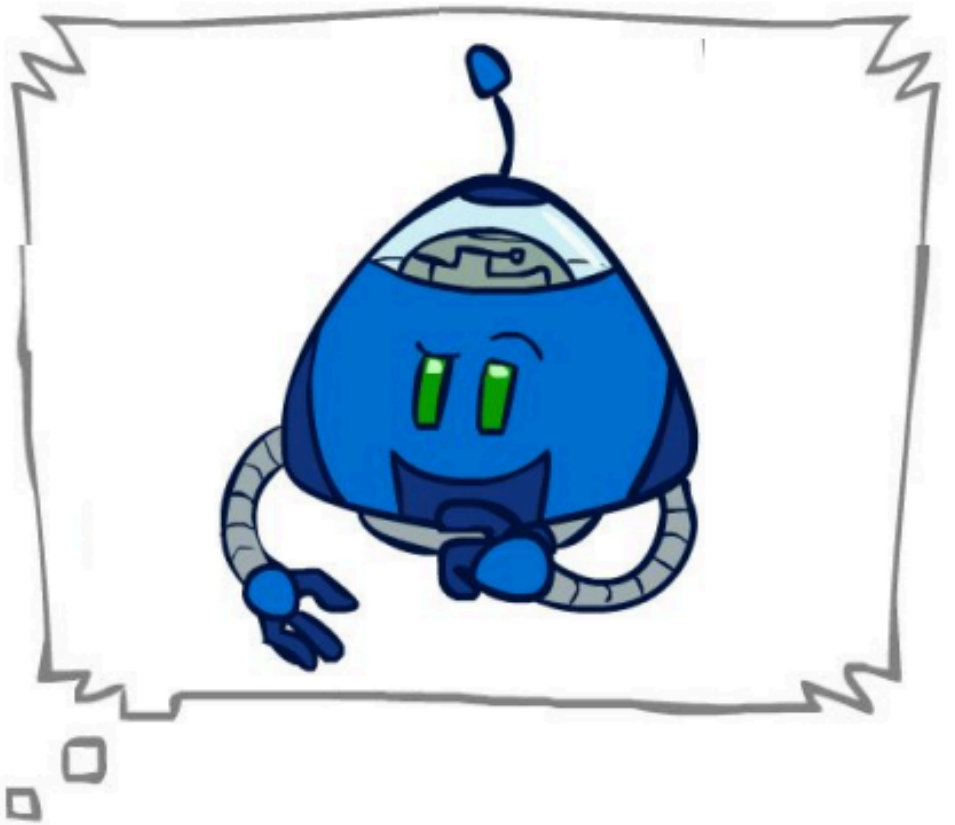
- Reinforcement learning

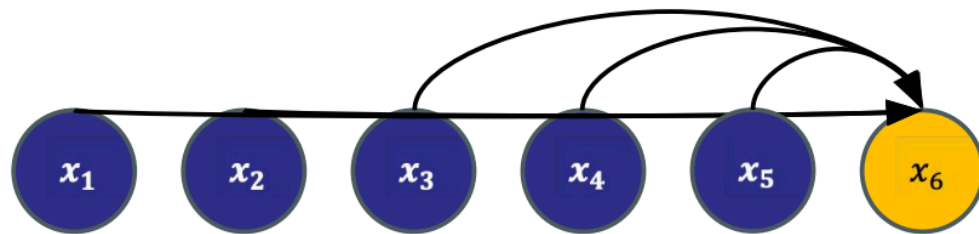
- ... from human feedback (RLHF)



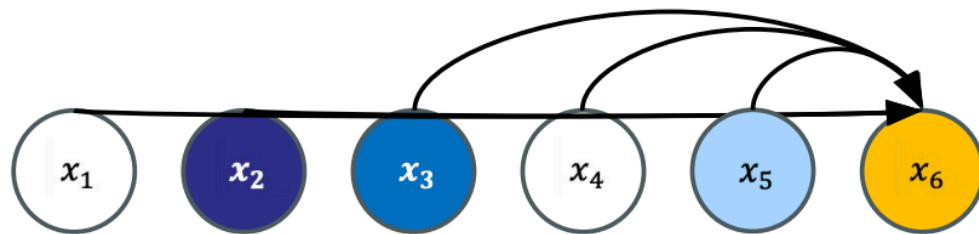
- Predict output one piece at a time (e.g. word, token, pixel, etc.)
- Concatenate: input + output
- Feed result back in as new input
- Repeat



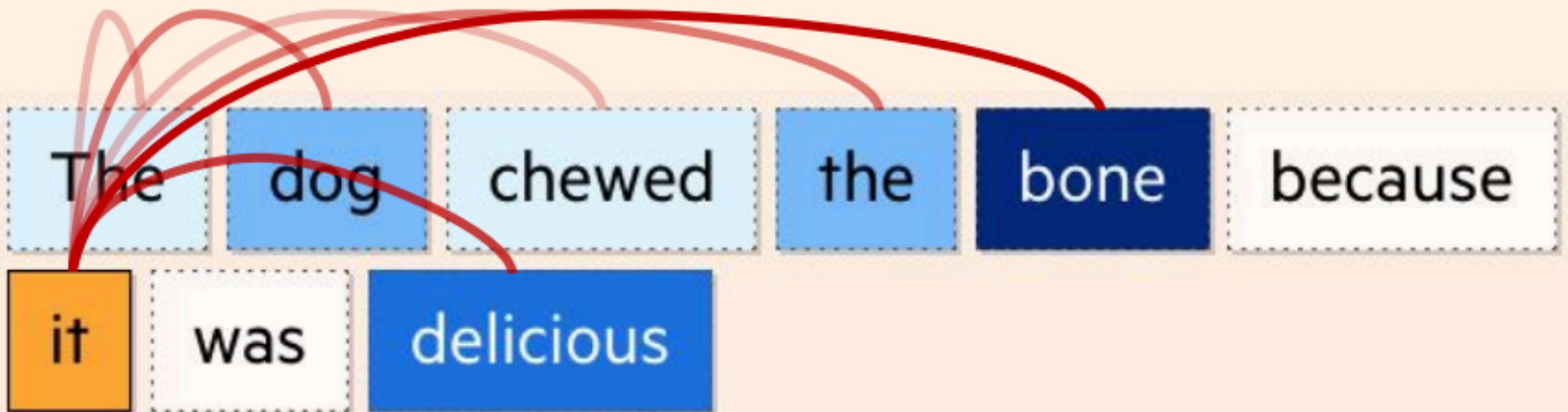
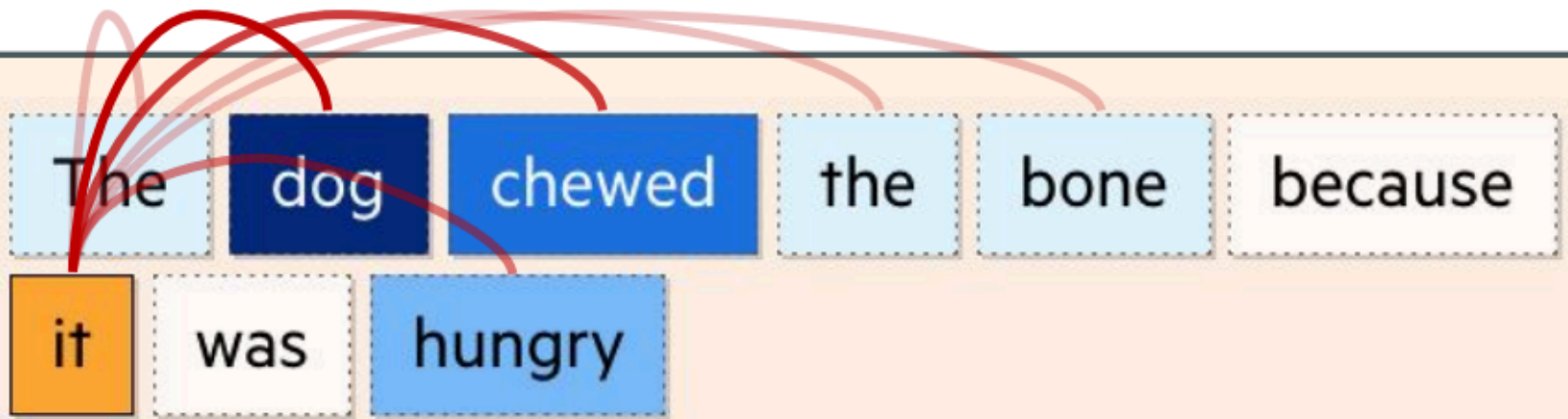


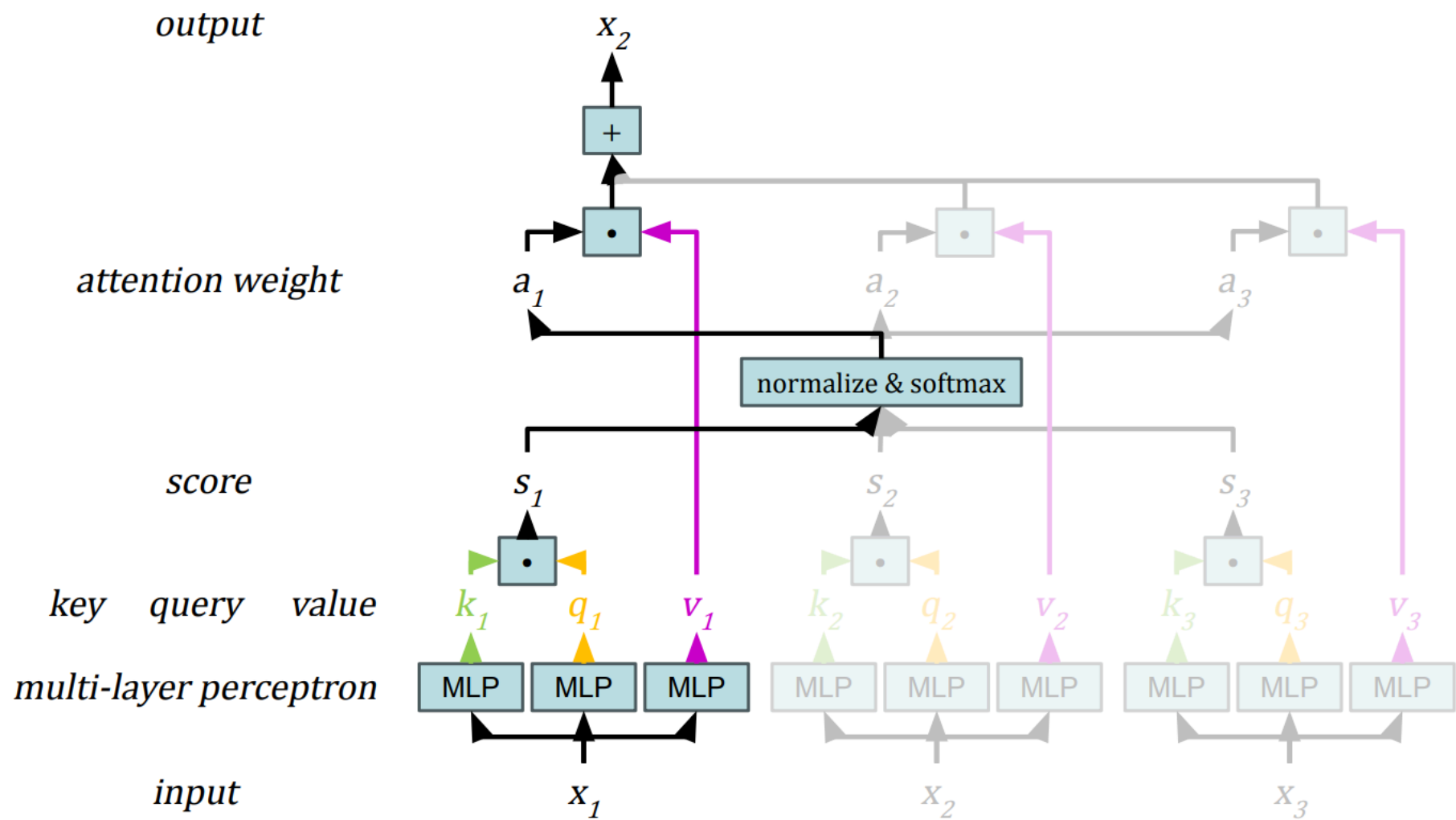


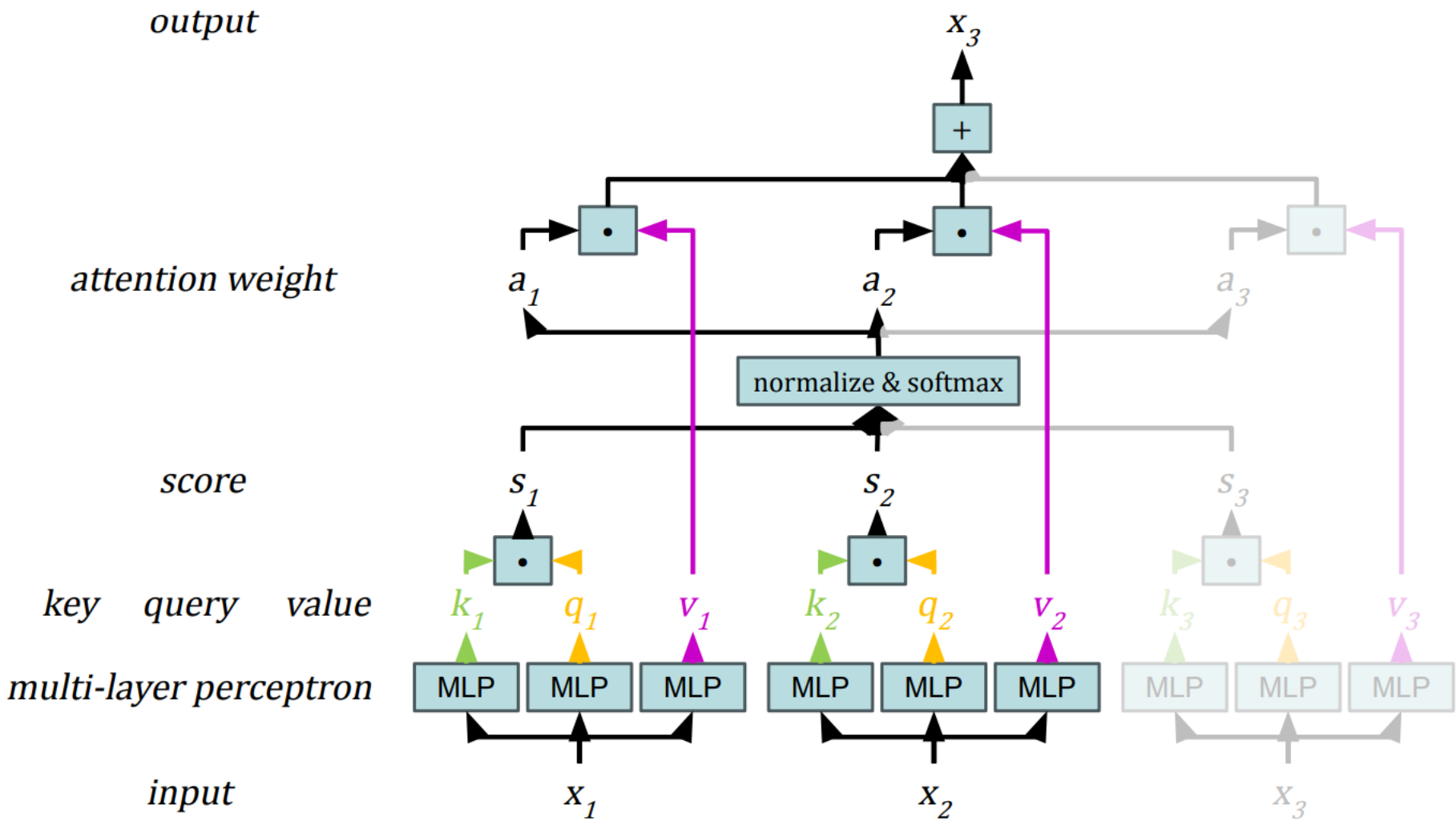
- Instead of conditioning on *all* input tokens equally...

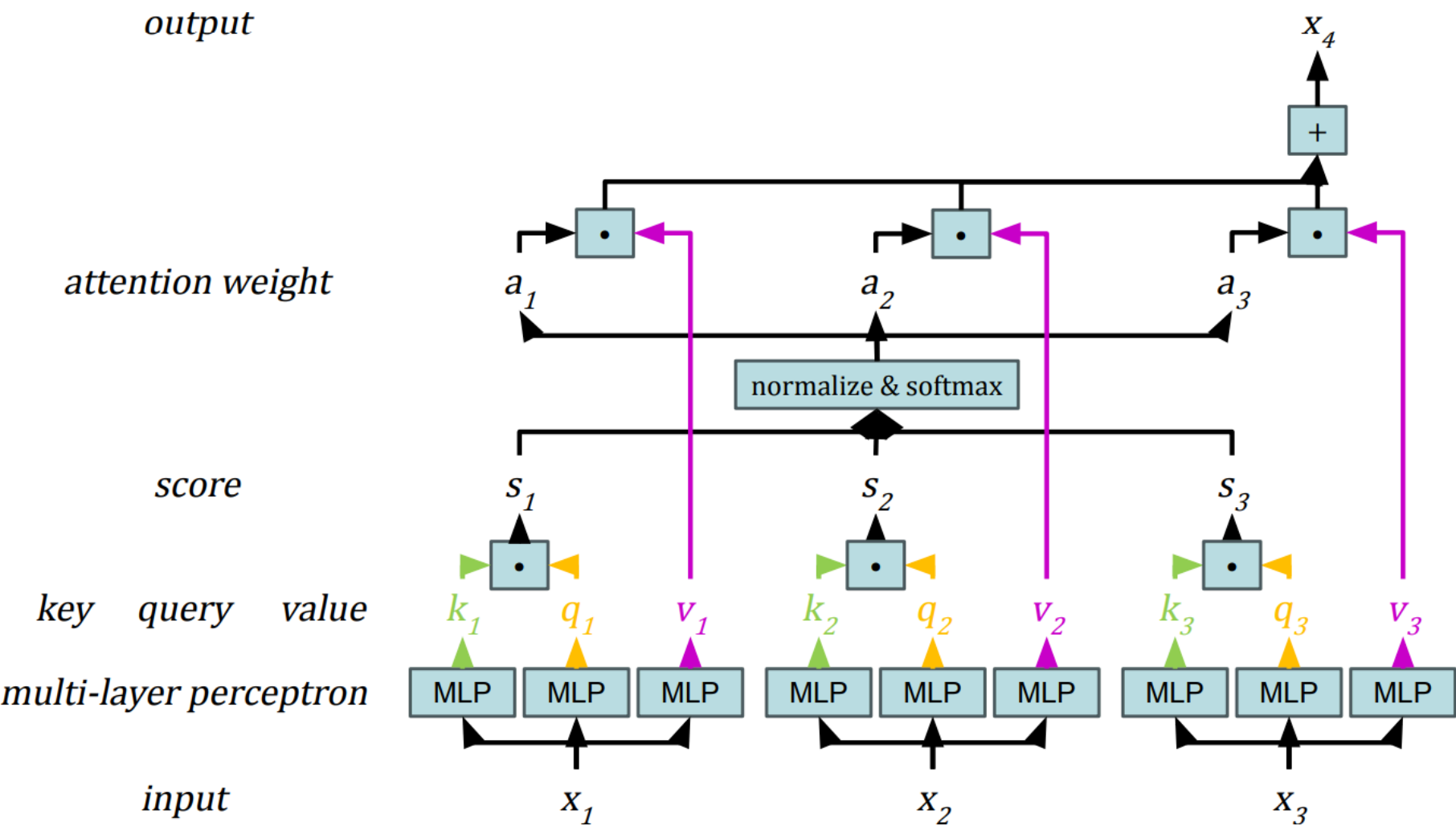


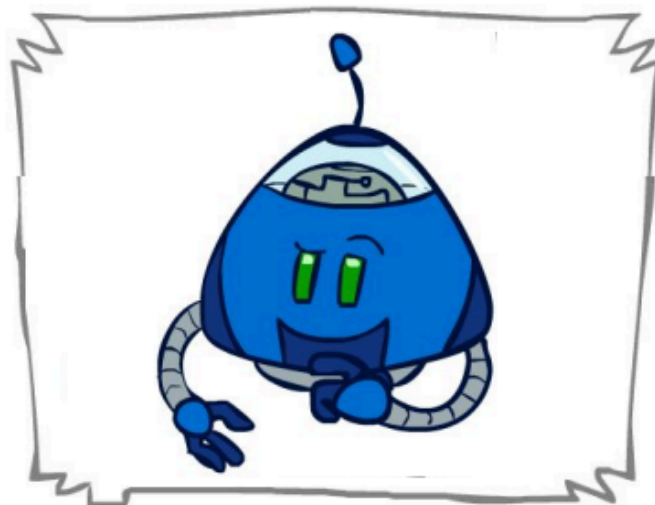
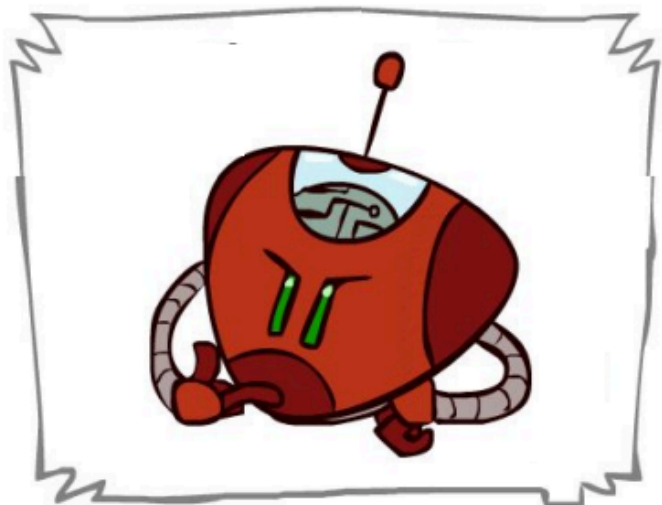
- Pay more attention to relevant tokens!



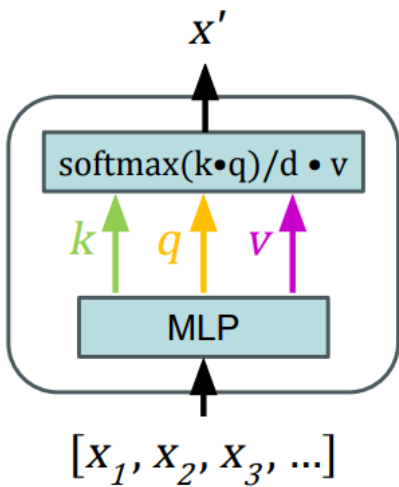




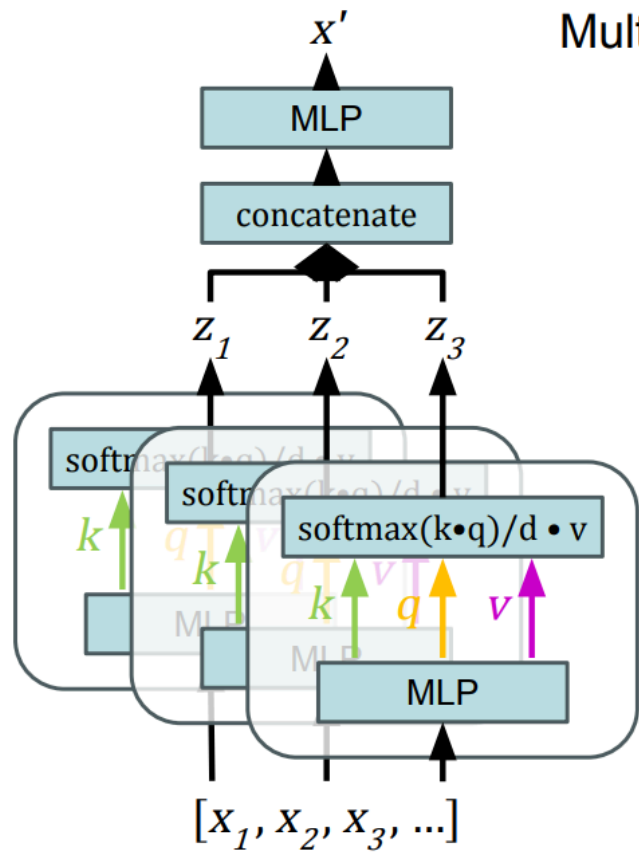




Single-headed



Multi-headed



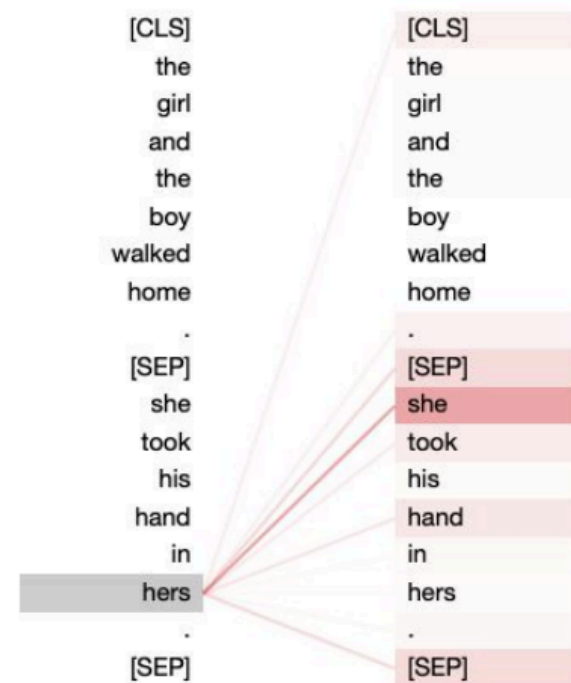
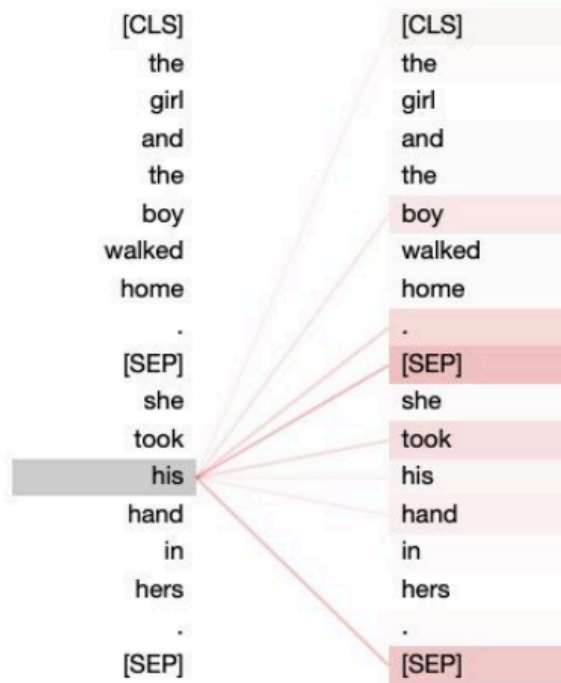
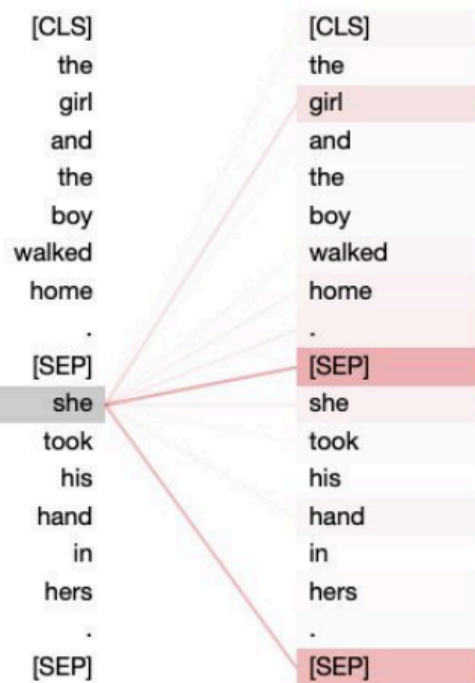
Head 6: previous word

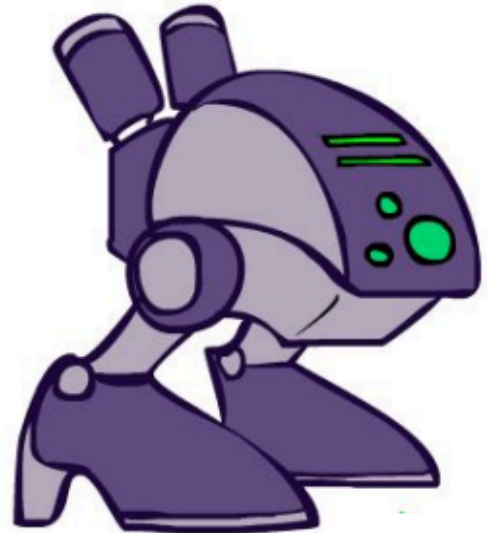
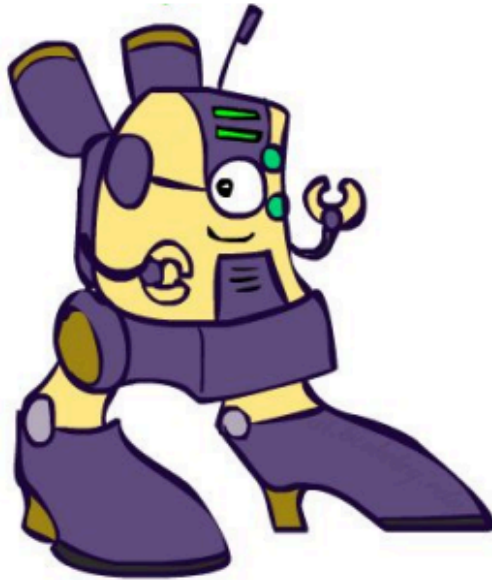
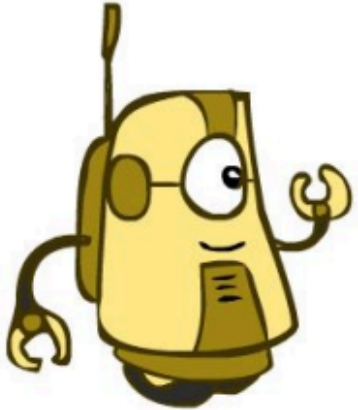
[CLS]	[CLS]
the	the
girl	girl
and	and
the	the
boy	boy
walked	walked
home	home
.	.
[SEP]	[SEP]

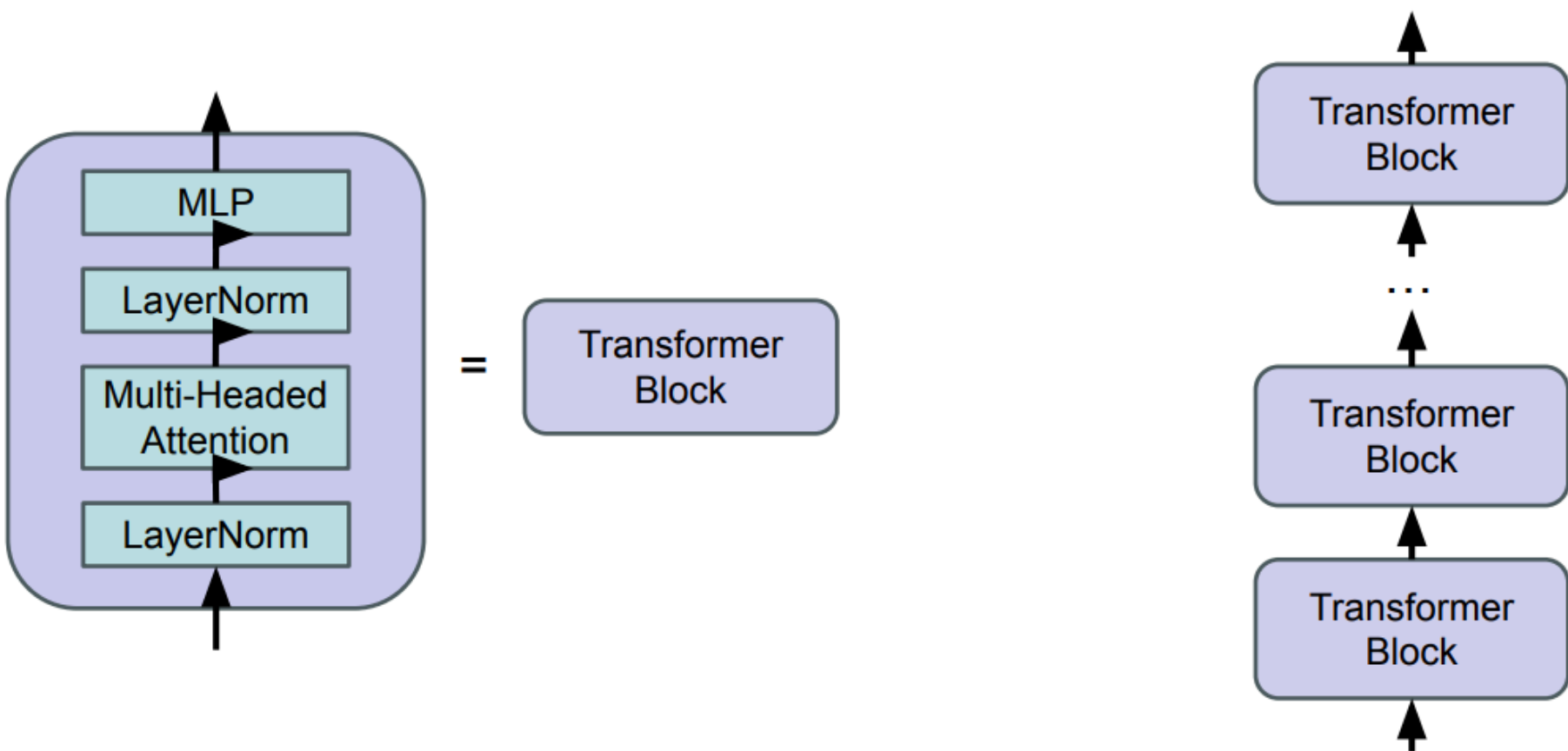
[CLS]	[CLS]
the	the
girl	girl
and	and
the	the
boy	boy
walked	walked
home	home
.	.
[SEP]	[SEP]

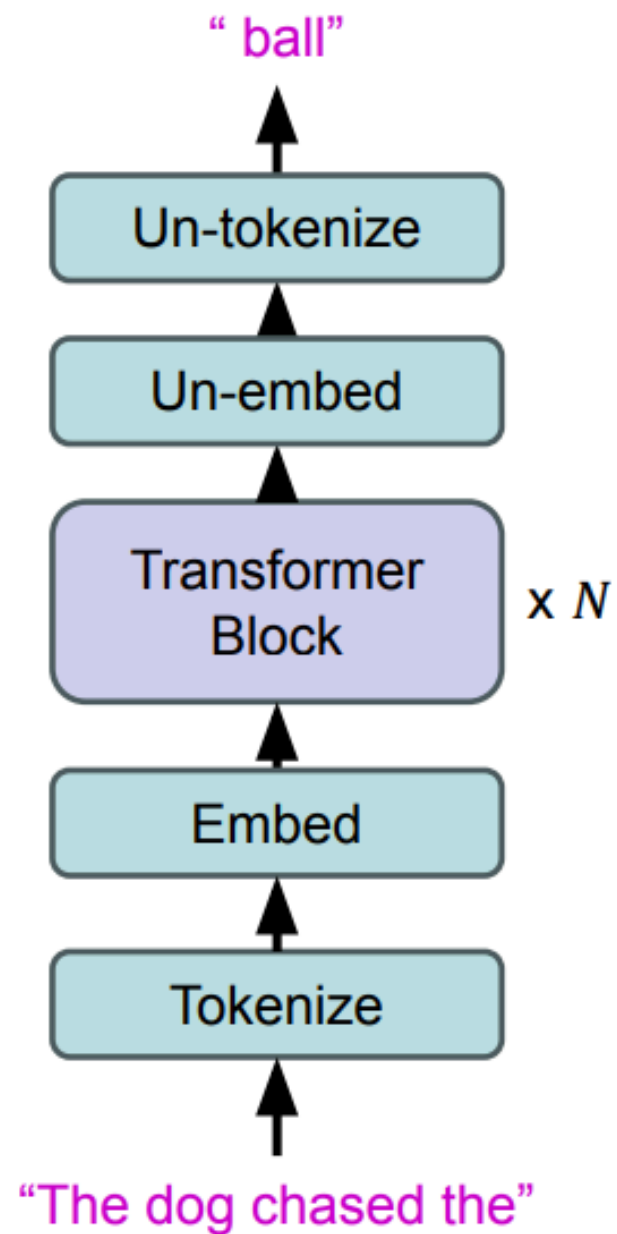
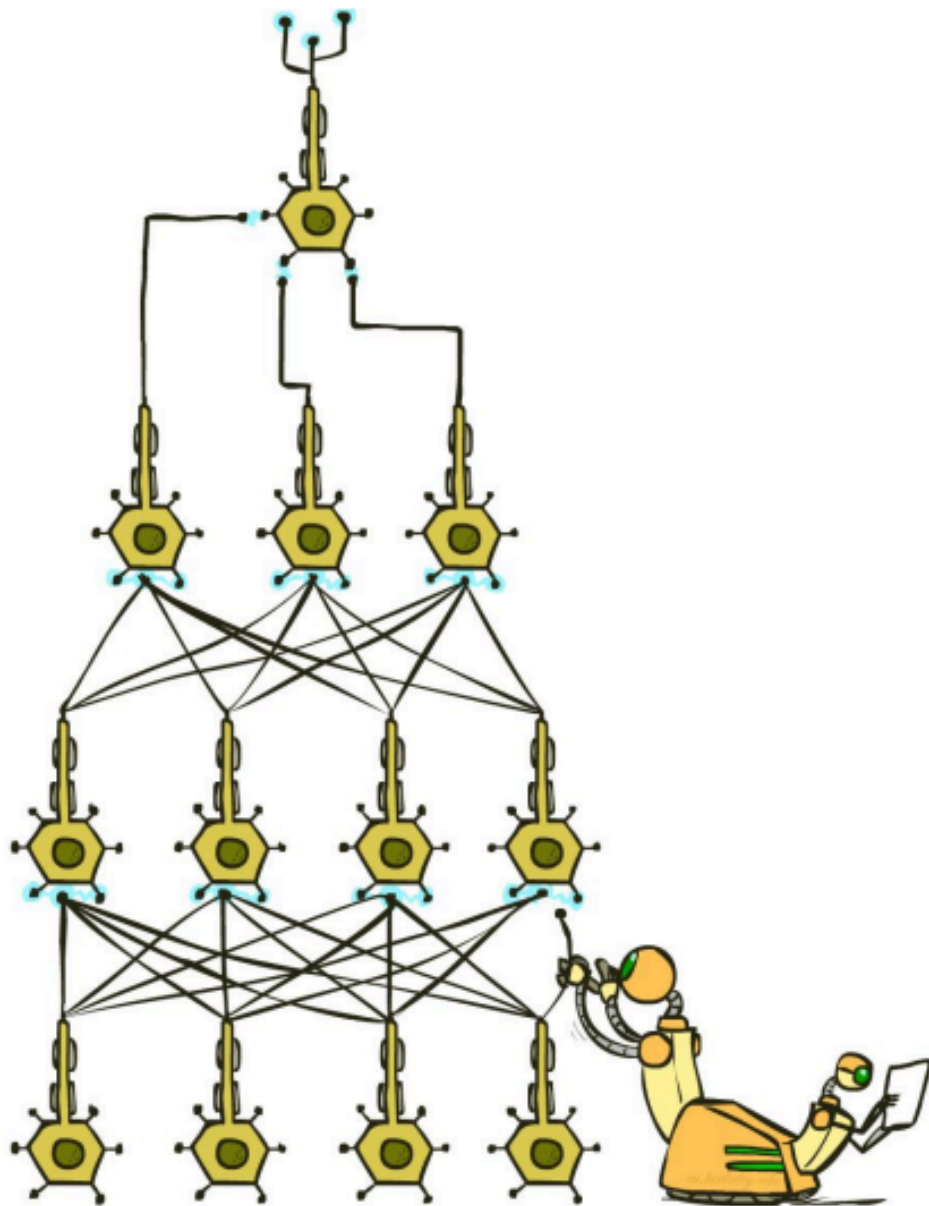
[CLS]	[CLS]
the	the
girl	girl
and	and
the	the
boy	boy
walked	walked
home	home
.	.
[SEP]	[SEP]

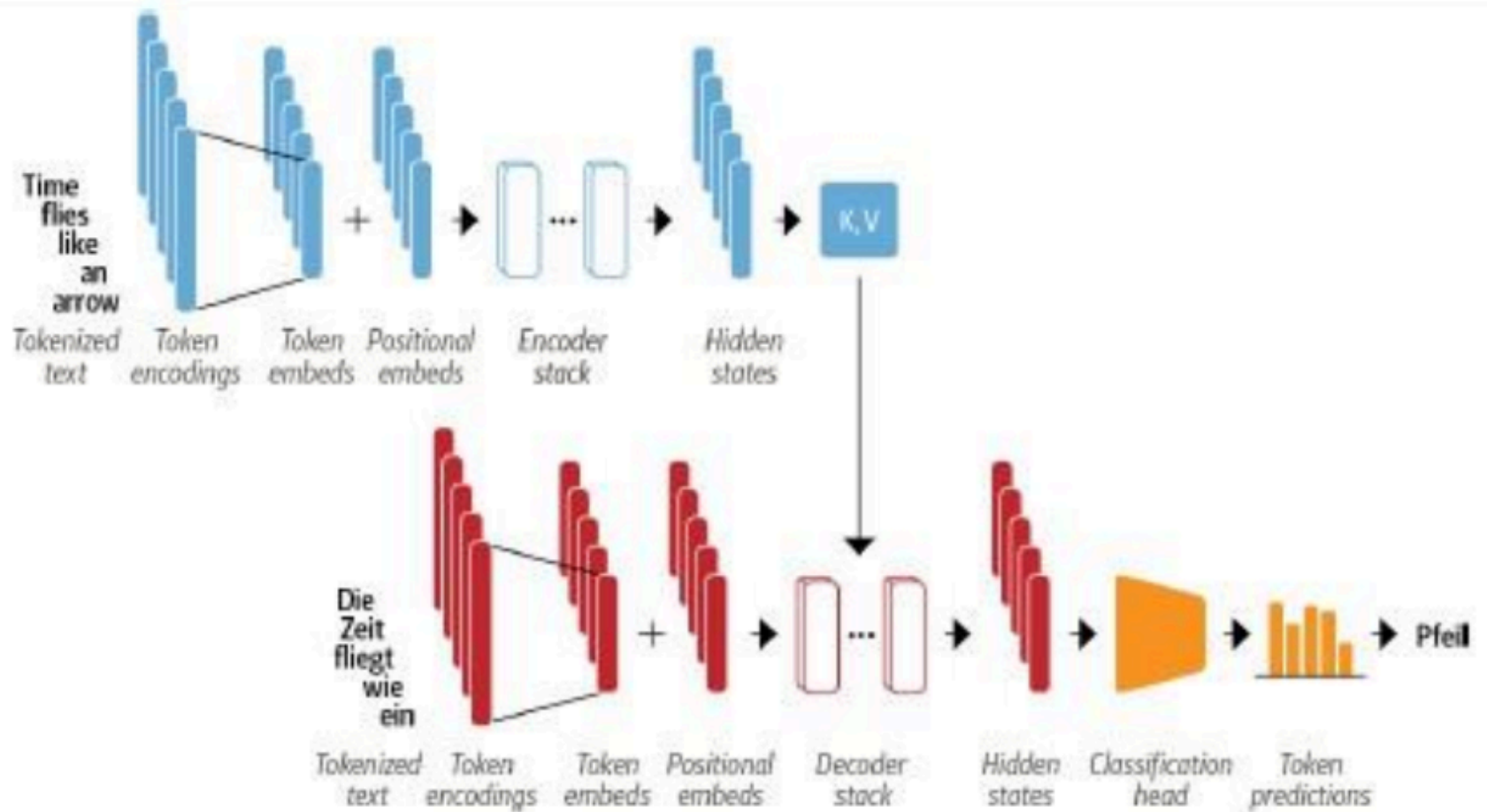
Head 4: pronoun references











■ ~~Feature engineering~~

- ~~Text tokenization~~
- ~~Word embeddings~~

■ ~~Deep neural networks~~

- ~~Autoregressive models~~
- ~~Self-attention mechanisms~~
- ~~Transformer architectures~~

■ ~~Multi-class classification~~

■ Supervised learning

- Self-supervised learning
- Instruction tuning

■ Reinforcement learning

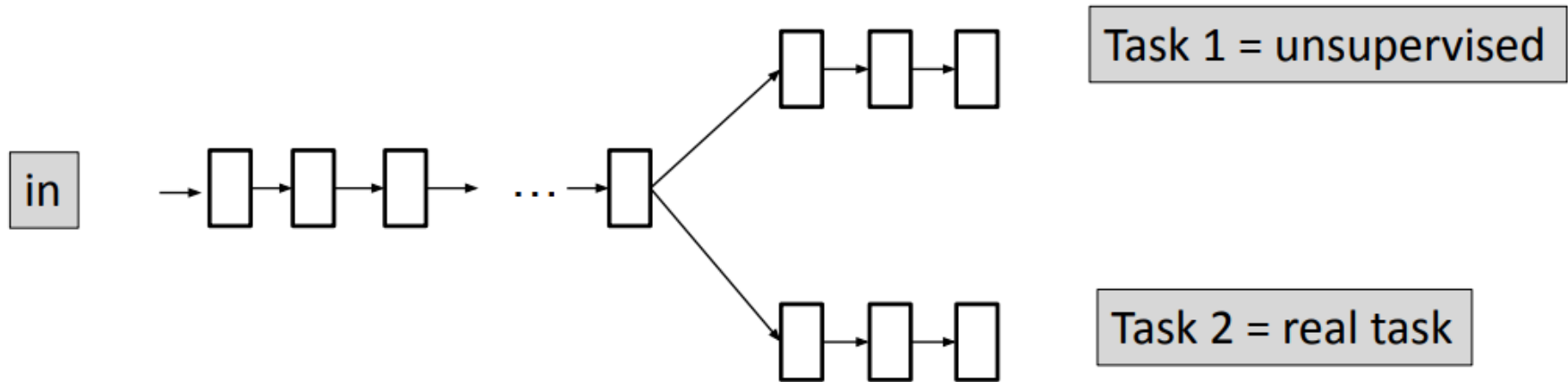
- ... from human feedback (RLHF)

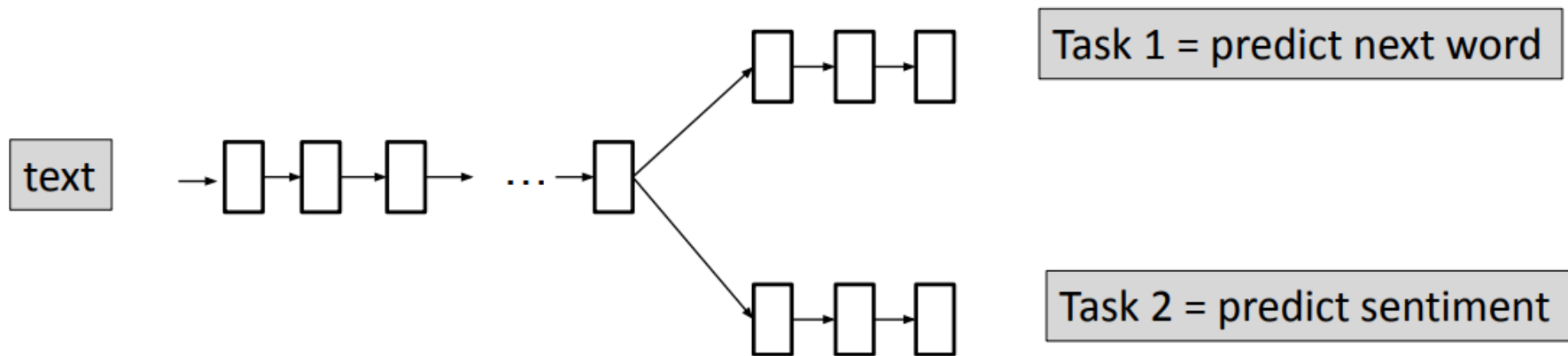
- Do we always need human supervision to learn features?
- Can't we learn general-purpose features?
- Key hypothesis:

Task 1 IF neural network smart enough to predict:

- Next frame in video
- Next word in sentence
- Generate realistic images
- ``Translate'' images
- ...

Task 2 THEN same neural network is ready to do Supervised Learning from a very small data-set







1

Pre-Train: train a large model with a lot of data on a self-supervised task

- Predict next word / patch of image
- Predict missing word / patch of image
- Predict if two images are related (contrastive learning)

2

Fine-Tune: continue training the same model on task you care about

- Task 1 = predict next word (learns to mimic human-written text)
 - Query: "What is population of Berkeley?"
 - Human-like completion: "This question always fascinated me!"
- Task 2 = generate **helpful** text
 - Query: "What is population of Berkeley?"
 - Helpful completion: "It is 117,145 as of 2021 census."
- Fine-tune on collected examples of helpful human conversations
- Also can use Reinforcement Learning

■ MDP:

- **State:** sequence of words seen so far (ex. "What is population of Berkeley? ")
 - 100,000^{1,000} possible states
 - Huge, but can be processed with feature vectors or neural networks
- **Action:** next word (ex. "It", "chair", "purple", ...) (so 100,000 actions)
 - Hard to compute $\max_a Q(s', a)$ when max is over 100K actions!
- **Transition T:** easy, just append action word to state words
 - s: "My name" a: "is" s': "My name is"
- **Reward R:** ???
 - Humans rate model completions (ex. "What is population of Berkeley? ")
 - "It is 117,145": +1 "It is 5": -1 "Destroy all humans": -1

- Clearly, we don't just want to predict the next word in internet text
- But even human feedback can have surprising / bad consequences

- Sycophancy
- Overconfidence
- Length
- ...



- More generally: lots of bad things happen when there is a gap between *what we really want to optimize for* and *what we train the model to optimize for*

Desired target	Actual target	Bias
Patient health needs	Patient health costs	Disparities in access to care
Severity of knee osteoarthritis	Severity as assessed by radiologist	Radiologists overlook features affecting underserved populations
Crime rates	Arrest rates	Disparities in policing
What users value	What users click on	Clickbait

river

water

ocean

upward

airplane

sign

traffic

heel

head

toe

stop

go

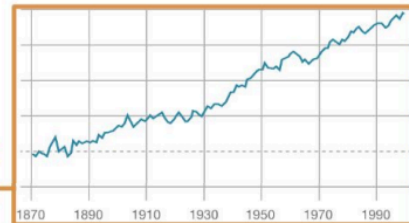


river

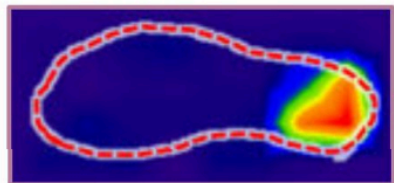
water

ocean

upward



airplane



heel

head

toe

sign traffic

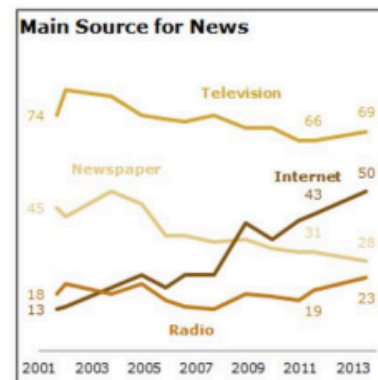


stop

go



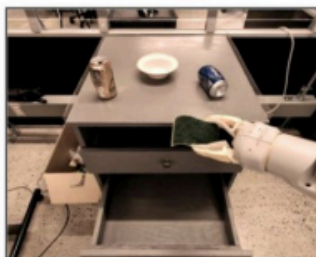
was invented by Wright brothers. Who invented
example from [Tsimpoukelli et al, 2021]



What is the fastest-growing news source according to

?

If  changes into  what does  change into?



What action should I take from

to accomplish "



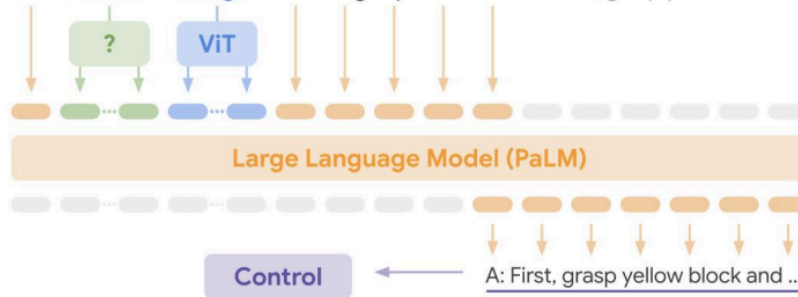
"?

Visual Q&A, Captioning ...



A:       .

Given `<emb>` ... `<img>` Q: How to grasp blue block? A: First, grasp yellow block



A dog jumping over a hurdle at a dog show.

Q: Miami Beach borders which ocean? A: Atlantic. Q: What is 372×18 ? A: 6696. Q: Write a Haiku about embodied LLMs. A: Embodied language. Models learn to understand.
The world around them.

Given **<emb>** Q: How to grasp blue block?
A: First grasp yellow block and place it on the table, then grasp the blue block.

Given  Task: Sort colors into corners.

Step 1. Push the green star to the bottom left.

Step 2. Push the green circle to the green star.